

Total No. of Questions : 8]

SEAT No. :

PD-4107

[Total No. of Pages : 2

[6402]-67

**S.E. (Artificial Intelligence & Machine Learning)**  
**OBJECT ORIENTED PROGRAMMING**  
**(218544) (2019 Pattern) (Semester - III)**

Time : 2½ Hours]

[Max. Marks : 70

Instructions to the candidate:

- 1) Answer Q1 or Q2, Q3 or Q4, Q5 or Q6, Q7 or Q8.
- 2) Neat diagrams must be drawn wherever necessary.
- 3) Figures to the right side indicate full marks.
- 4) Assume Suitable data if necessary.

- Q1)** a) What is Destructor? Explain the use of destructor in object oriented programming. [6]
- b) Discuss the advantages and disadvantages of constructor overloading. [6]
- c) Explain different types of Constructors in Object Oriented Programming. [6]

OR

- Q2)** a) Differentiate constructors from regular methods in object-oriented programming, considering their purpose, invocation, return type and inheritance properties. [9]
- b) Compare and contrast different types of constructors, such as default constructors, parameterized constructors and copy constructors. [9]
- Q3)** a) Differentiate between abstract classes and interfaces in object-oriented programming. [5]
- b) Differentiate between compile-time polymorphism and runtime polymorphism. [6]
- c) Explain how inheritance can be used to model real-world relationships between entities. [6]

OR

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- Q4)** a) Identify and explain the different types of inheritance, such as single inheritance, multiple inheritance and multilevel inheritance. [8]
- b) Analyze the impact of polymorphism on the efficiency of a software system, considering both compile-time and run-time aspects. [9]
- Q5)** a) Explain the concept of an exception in object-oriented programming. [6]
- b) Provide examples of each type of error and how they manifest in programming code. [6]
- c) Discuss the ArrayList class and how it implements the List interface. [6]

OR

- Q6)** a) Develop a simple generic class that can work with different data types. Provide an example of how this class can be used in a program. [9]
- b) Discuss the consequences of uncaught exceptions and their impact on program execution also discuss purpose and use of multiple catch clauses in a try-catch block. [9]
- Q7)** a) What are the different stream classes and what is the purpose of each stream class? [8]
- b) Define design patterns and discuss three main categories of design patterns? [9]

OR

- Q8)** a) Analyze the strengths and weaknesses of different design patterns. [8]
- b) Explain the concept of random access files and provide examples of applications that use random access files. [9]

